Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the accuracy during training.

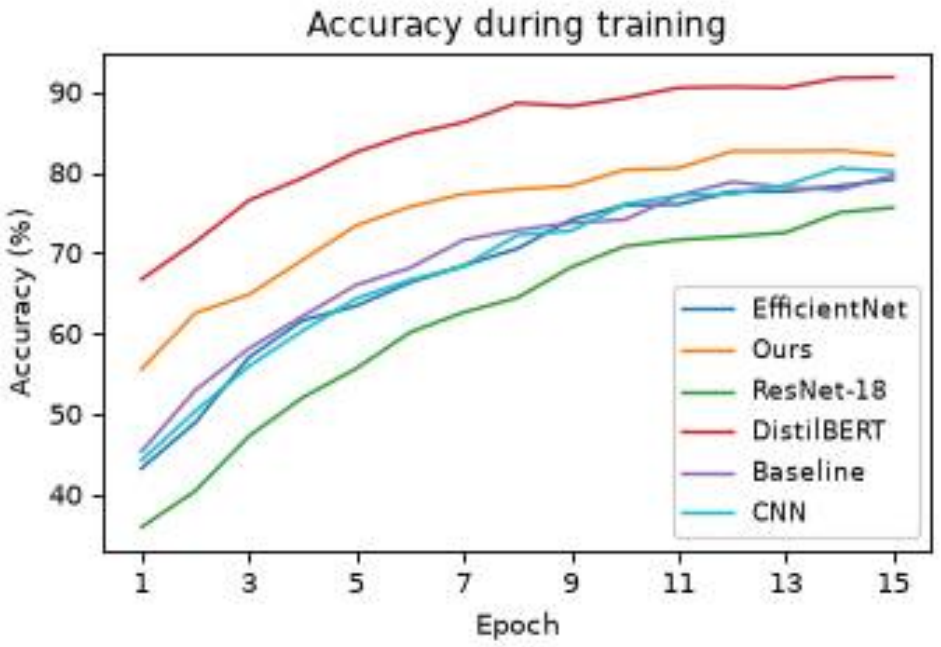


Figure 1: Accuracy during training for the models in this experiment.

EfficientNet reaches 72.2% accuracy at epoch 14. As shown in Figure 1, at epoch 10, DistilBERT is 15.3 points better than ResNet-18. Ours reaches 75.7% accuracy at epoch 6. From epoch 11 to epoch 12, the accuracy of CNN improves by a relative 40.1% (Figure 1).

Figure 2 shows the test accuracy during training.

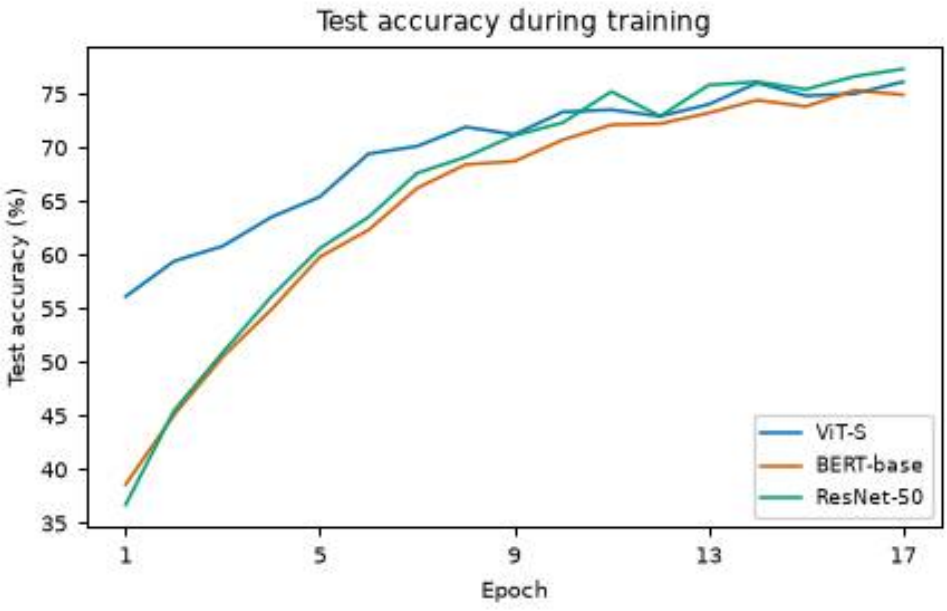


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 6, ResNet-50 achieves 63.5% test accuracy. At epoch 16, ResNet-50 achieves over 75% test accuracy.

Figure 3 shows the f1 score during training.

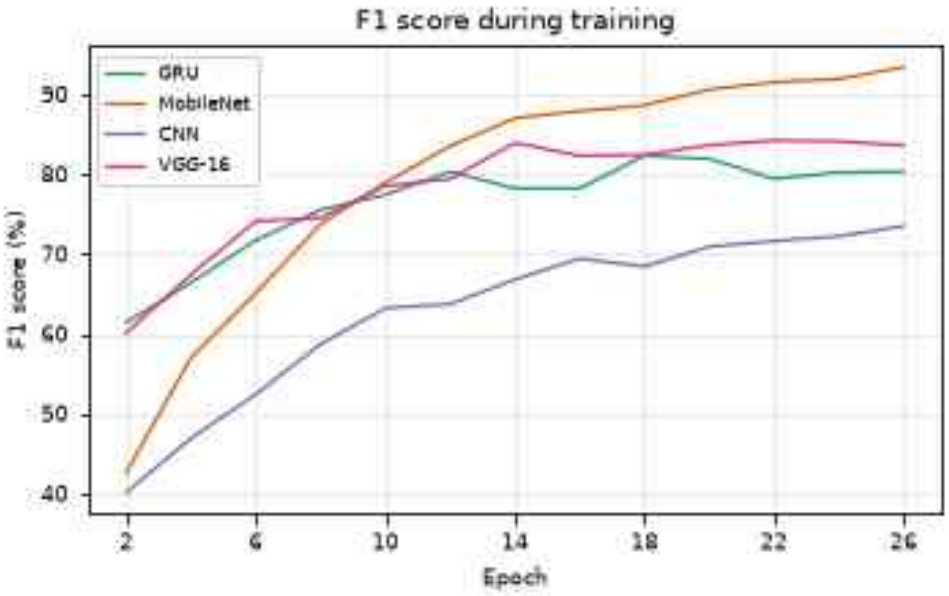


Figure 3: F1 score during training for the models in this experiment.

As shown in Figure 3, at epoch 16, MobileNet achieves over 85% F1 score. CNN attains a F1 score of 74.7% at epoch 26. As shown in Figure 3, at epoch 6, GRU is 19.3 points better than CNN.

Figure 4 shows the validation accuracy of GRU by dataset.

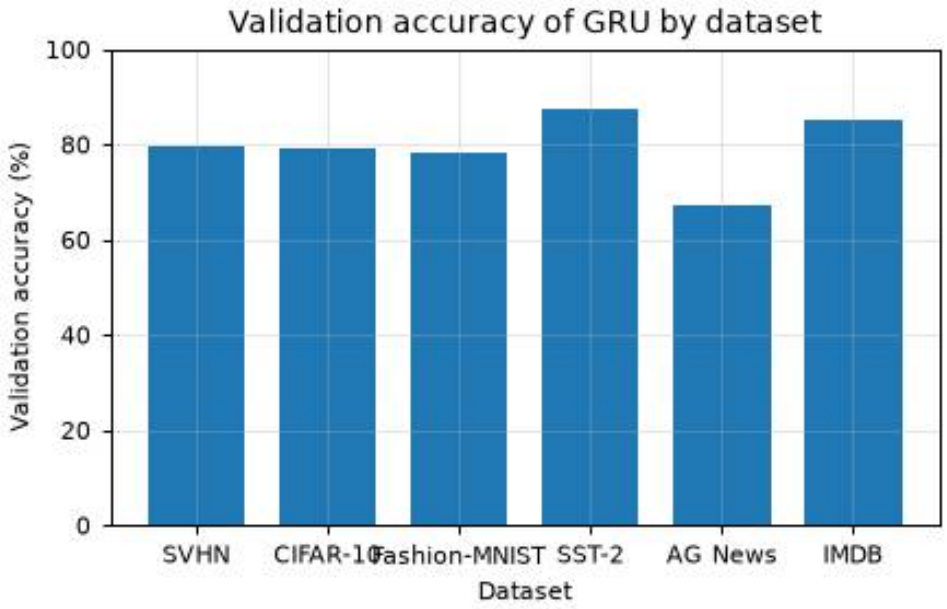


Figure 4: Validation accuracy of GRU by dataset for the models in this experiment.

GRU scores 6 points higher on IMDB than on SVHN. GRU reaches 87.4% validation accuracy on SST-2 (Figure 4). As shown in Figure 4, GRU reaches over 75% validation accuracy on CIFAR-10. On IMDB, GRU obtains a validation accuracy of 85.3% (Figure 4).

3 Discussion

SVM reaches 54.7% accuracy on CIFAR-10. The differences between the models are discussed in the next section.